

Analysis Geometrica Propria

Properly Geometric Analysis

G. W. Leibniz, 1698

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We will attempt here, as something of a sample, *properly geometric analysis* and the *calculus of situation* connected with it, lest by chance the thought is lost, which has not come to mind for anyone else, as far as I know, and which will provide very different applications from those furnished by algebra. What needs to be understood is that magnitude and situation are different considerations. There is magnitude even in things that do not have situation, such as number, proportion, time, velocity; namely wherever there exist parts which can be assessed through number, i.e. by repetition. Therefore the doctrines of magnitudes and of numbers are the same; and algebra itself, or logistic (if you prefer), when treating of magnitude in general, in fact deals with unspecified, or at least unnamed, number. But situation adds a certain form over and above the magnitude or the multitude of parts, as in figurate numbers. Hence, algebra clearly comprises analysis properly and in itself belonging to arithmetic, even if it is transferred to geometry and situation to the extent that it handles the magnitudes of curves and figures. But then algebra necessarily supposes much that is proper to geometry or situation which should be released from it. Our analysis, therefore, effects this release, assuming or supposing nothing more, and being not so much accommodated to magnitude as to situation itself. Moreover, for explaining the matter of situation we now use nothing but *congruence*, reserving *similarity* and *motion* to another place.

(1) Things are *Congruent* which can be substituted for one another in the same place, as ABC and CDA [Figure 1], which I signify thus: $ABC \simeq CDA$. For $\sim$ is my sign of similarity, and $=$ of equality, from which I compose the sign of congruence, since things that are simultaneously similar and equal are congruent.

(2) Things are counted as *relating in the same manner*, or *congru-*

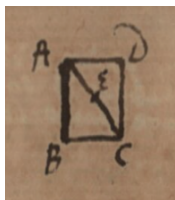


Figure 1

ently, which correspond to each other in congruent things. For example, $AB.ABC \simeq CD.CDA$, namely AB is to ABC as CD to CDA , and so $AB.AC \simeq CD.CA$ and $A.BC \simeq C.DA$, that is, the point A relates in the same manner to the line BC as the point C to DA . Indeed here it is done not just by ratio or proportion, but by whatever relation.

(3) *Axiom*: If the determiners are congruent, so too will be the things determined, supposing of course the same way of determining. For example, if $A.B.C \simeq D.E.F$, then the circumference of the circle through $A.B.C$ is also congruent to the circumference of the circle through $D.E.F$, since the circumference of a circle is determined by three given points. And in general, in the calculation of congruents, determined things may be substituted for sufficient determiners, just as equals are substituted for equals in common calculation. See below §26, §30, §32.

(4) In order to understand the calculus better, it should be observed that when we say $A.B.C \simeq D.E.F$, it is the same as if we had said simultaneously: $A.B \simeq D.E$ and $B.C \simeq E.F$ and $A.C \simeq D.F$, so that these can be made from that by *disjoining*, and that from these by *combining*. See below §26, §27, §29, §30, §31, §32.

(5) A *common boundary* [*terminus*] is a locus which is in [*inest*¹] two loci, such that it is not a part of them. In this way a point E is a locus which is in the lines AE , CE , but is not part of either. Therefore it is

¹Leibniz is probably using *inest* in his technical sense associated with his mereological calculus: X *inest* Y when, X being supposed, Y is immediately thereby also supposed. We do not flag subsequent instances in this essay.

said to be their common boundary.

(6) A *section* is the entire common boundary of two parts constituting a whole and not having a common part. Thus AC is the entire common boundary of triangles ABC , CDA constituting the whole $ABCD$ and not having a common part.

(7) We will express this by the calculus, through which geometry is carried back to logic, in this way: Every point that exists in a proposed locus is designated by a common mark or letter (for example) X , and the locus itself is designated by $\overline{X}$, by drawing a line over the letter. If some given points of the locus are Y and Z , [their] loca will be $\overline{Y}$ or $\overline{Z}$. Let, therefore, the whole be $\overline{X}$, the constituting parts be $\overline{Y}$ and $\overline{Z}$, and $\overline{V}$ be the section; then these propositions can be formed: Every Y is X , every Z is X , since $\overline{Y}$ and $\overline{Z}$ are in $\overline{X}$. But also, whatever is neither Y nor Z is not X , having set $\overline{Y}$ and $\overline{Z}$ to be *parts constituting*, or *exhausting*, the whole $\overline{X}$. Further, every V is Y , and every V is Z , since $\overline{V}$ is in *common* to $\overline{Y}$ and $\overline{Z}$, i.e. is in both of them. Finally, whatever is simultaneously Y and Z is also V , since $\overline{V}$ is the *section*, i.e. the entire common boundary, namely that which contains what is common to both, and indeed they don't have a common part (i.e. something besides a boundary). Hence all subalterns, conversions, oppositions, and consequences of logic have a place here, sometimes fruitfully, although other times they seem to be precluded from real [use] by human weakness, not their own fault.

(8) The loci $\overline{X}$ and $\overline{Y}$ *coincide* if every X is Y and every Y is X . I signify it thus: $\overline{X} \approx \overline{Y}$.

(9) A *point* is a locus in which no other locus can be taken; and so if the locus $\mathfrak{D}$ is taken in the point $\odot$, then $\mathfrak{D}$ coincides with $\odot$, and if in turn $\mathfrak{D}$ is in $\odot$ and from this alone one may conclude that $\odot$ and $\mathfrak{D}$ coincide, then $\odot$ is a point.

Absolute space is the opposite of a point, for in space every other locus can be taken, as none can be taken in a point, so the point is the simplest with respect to situation, and as it were the minimum, whereas space is the most diffuse, and as it were the maximum.

(10) A *body* (mathematical, of course), or *solid*, is a locus in which

there is more than boundaries. And this is what we mean when we attribute depth to a solid. Conversely, whatever is in a surface or curve can be understood to be a boundary, and is in common to something together with another thing not having a common part with it. There is an analogy between point and solid here also, that whatever is in a point is a point, and conversely whatever a solid is in, is a solid. Likewise, a point cannot be in something as a part; but a solid cannot be in anything other than as a part.

(11) A *plane* is a section of a solid relating in the same way on both sides² to what does not touch the boundaries of the solid, i.e. relating in the same way on both sides to what ends up on one side as to what ends up on the other. If you cut an apple by a plane, the faces [*extrema*] of the two pieces, by which they were attached, cannot be distinguished from each other.

(12) And so if it is an unbounded solid, it is absolutely true that the dividing plane relates in the same way to both sides. But if the solid is bounded, it is enough that the boundaries do not enter into consideration. And something done the same way on both sides will indeed relate in the same way to the section.

(13) A *line* is a section of a plane relating in the same way on both sides to what does not touch the boundaries of the plane.³

(14) Let there be an unbounded plane AA [Figure 2], and its sec-



Figure 2

²'Relating in the same way' is our fixed expression for *eodem modo se habens*. Here and below it is joined with *utrinque*, 'on or from both sides or parts'.

³Leibniz added the following marginal remark: 'What if someone doubted whether the plane could be so divided? Is it preferable, therefore, to form a line as the intersection of two planes?'

tion BB relates in the same way on both sides; then BB will be an unbounded line.

(15) But if a plane CC is bounded [Figure 3], no matter what its

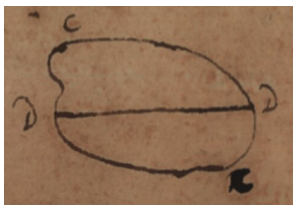


Figure 3

shape, if we nonetheless cover the boundaries so that they don't appear, or we in no way take them into account, then we will discover a dividing line DD to relate in the same way on both sides, and it will be bounded.

(16) But a curve relates differently on each side, since it is concave on one side, and convex on the other.

Now, everything that follows should be understood to be in a plane.⁴

(17) If there is a line [Figure 4] in which there are points A and B ,

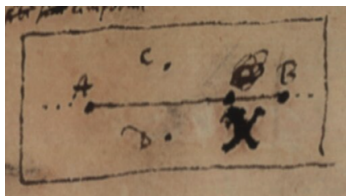


Figure 4

and outside it a point C on one side, then necessarily another D can be given on the other side, which relates in the same way to A and B as C does to them. For otherwise, since these points are in the line by hypothesis, one side would not relate to the line like the other, contrary

⁴Leibniz's marginal note: 'It seems necessary for some property of the plane to enter the argument, such as that two planar figures with the same boundary are congruent, i.e. that the interior is uniform.'

to the definition of a line. Therefore, given $C.A.B$, one can find D such that $C.A.B \simeq D.A.B$.

(18) Therefore if a point X is given, unique with its relation to two points $A.B$, it could not be on either side of the line through $A.B$, otherwise another one would be given by the preceding, its twin, contrary to hypothesis. Hence it is necessary that it falls on that very line, where what are elsewhere twins merge into one, since the line is also the common boundary of both sides.⁵

(19) Consequently, a straight line (bounded, of course) is the locus of all points unique with their relation to two points in itself. If $X.A.B \simeq Z.A.B$, and for that reason X coincides with Z , then $\overline{X}$ is the (unbounded) line through $A.B$.⁶

(20) From this one now concludes that two lines could not meet each other except in one point, i.e. two lines which have two points A and B in common coincide with each other when extended, since either one is the locus of all points (and thus of the same ones) unique with their relation to points A and B . And so, given two points, a line is determined on which they fall.

(21) Hence, moreover, two lines, namely which do not coincide when extended, cannot have a segment in common. For if they have a segment AB in common [Figure 5], then they also have at least the two points $A.B$ common; therefore [when] extended they coincide.

(22) Similarly, two lines cannot enclose space, otherwise they will meet each other twice, and thus they will have two common points [Fig-

⁵Leibniz's marginal note: 'The converse still needs to be demonstrated, that every point in a line is unique with its relation to two points in it.'

In turn, every point in the line through $A.B$ is unique with its relation to those two points. Let such a point be X ; if it is not unique, then an Ω is given which is to $A.B$ as X is to $A.B$; since X is in the line through $A.B$ by hypothesis, then so too is Ω ; there is therefore some order among the four points $A.B.X\Omega$ in the straight line, therefore $A.B.X$ and $A.B.\Omega$ are not $\simeq$. Supposing that the locus is a *curve*, there is not an order of all points in the (linear?) curve. Therefore there are two points in it relating in the same way to two points in itself, and so the locus determined with respect to the two is a curve.'

⁶Leibniz's marginal note: 'In general, every point in a curve not returning to itself is unique with its relation to two points taken in it.'

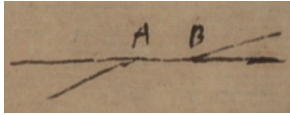


Figure 5

ure 6].



Figure 6

(23) Thus, from our definition of the line we demonstrate axioms which Euclid assumes about the line without demonstration.

(24) A *circle* is produced by the motion of a line around a single point remaining stationary in the plane. The stationary extremum is the centre, and the curve described by the other extremum is the circumference.

(25) And so [Figure 7] every point of the circumference, say X , will

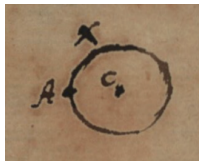


Figure 7

relate in the same way to the centre C , i.e. every X will relate to C as A does to C . As expressed in our calculus, this is: if $X.C \simeq A.C$, then $\overline{X}$ will be the *circumference* of the *circle*.

(26) The locus of all points relating in the same way to two points is a *straight line*. Let there be two points C and D [Figure 8], and let there be a locus $\overline{X}$ any point X of which relates in the same way to C as to D ; I claim that $\overline{X}$ is a line. In order to demonstrate this, let two points

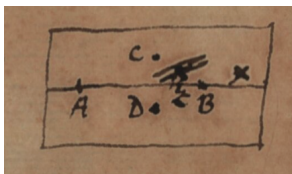


Figure 8

A and B be taken in the locus $\overline{X}$, and the line $\overline{Z}$ be drawn through A and B ; it is determined from $A.B$ by §20. Now $A.B.C \simeq A.B.D$ by hypothesis, since A and B fall in $\overline{X}$; therefore (by an axiom in §3) the line through $A.B$, or $\overline{Z}$, will also relate in the same way to C as to D , or $\overline{Z}.C \simeq \overline{Z}.D$; and now too $X.C \simeq X.D$ by hypothesis, therefore by combining, $X.\overline{Z}.C \simeq X.\overline{Z}.D$. Therefore the point X cannot be on one side of the line $\overline{Z}$, such as (if you will) on the D side; for then it would relate differently to the line $\overline{Z}$ and to D than it does to the line $\overline{Z}$ and to C ; and so it is necessary that X falls in $\overline{Z}$, i.e. every X will be Z , whence also $\overline{X}$ falls in $\overline{Z}$, *quod erat demonstrandum*.

(27) So we have here an elegant sample of the calculus following the rule of §4. Namely, since $X.\overline{Z} \simeq X.\overline{Z}$, which is identical, and $X.C \simeq X.D$ by hypothesis, and $\overline{Z}.C \simeq \overline{Z}.D$, which we proved from the nature of the line, from all these congruent pairs [*binionibus*] considered individually, it follows that the triples assembled from them are also congruent, i.e. that by combining,

$$\underbrace{X \cdot \overline{Z} \cdot C} \simeq \underbrace{X \cdot \overline{Z} \cdot D}$$

(28) Hence if $X.C \simeq X.D$, then $\overline{X}$ will be a *straight line*, which congruence is of the utmost utility in our calculus. And conversely, if $\overline{X}$ is a straight line, there should be points of such kind as C and D so a congruence has the locus.

(29) It cannot happen that a line relates in the same way to three points of the plane, i.e. that $X.C \simeq X.D \simeq X.E$ [Figure 9]. For if this were so, then by combining, $X.C.E \simeq X.D.E$, therefore E cannot be on

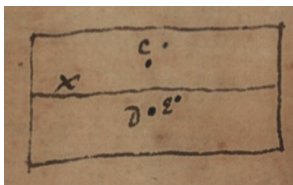


Figure 9

either side. But the same E cannot be in $\overline{X}$, for then C and D would also be in $\overline{X}$, and moreover, they would coincide with E , otherwise some point of the line (namely E itself) would relate differently to C and to D than to E , therefore a point E in addition to C and D cannot be found anywhere.

(30) A circle does not meet a circle in more than two points. Let two circular circumferences be $\overline{X}$ and $\overline{Z}$ [Figure 10]; I say they cannot cross

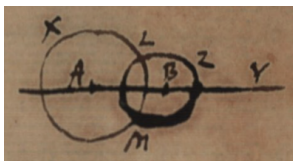


Figure 10

except in two points, such as L and M . Indeed, let the centre of $\overline{X}$ be A , the centre of $\overline{Z}$ be B . Now since L is X and M is X , $L.A \simeq M.A$, and since L is Z and M is Z , $L.B \simeq M.B$, both from the nature of the circumferences to which the points are common by §25. Therefore, $L.A.B \simeq M.A.B$ by combining. Suppose $\overline{Y}$ is the line through $A.B$, so that now $L.Y \simeq M.Y$ by the axiom of §3; but if, in addition, another point N is given, common to the two circumferences, we would have $L.Y \simeq M.Y \simeq N.Y$, or the line $\overline{Y}$ would relate in the same way to the three points L, M, N , which cannot happen by the preceding. Hence it follows that when three points are given, a circumference is determined which they are in [*insint*], since they cannot simultaneously be in multiple [circumferences].

(31) If a circle is tangent to a circle [Figure 11], the point of contact

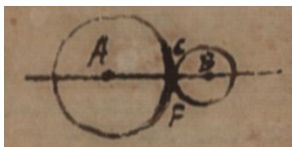


Figure 11

is in the same line as the centres. Let the centres be A and B , and the point of contact C , namely where two meeting points merge. And so (by the preceding) the circle does not meet the circle elsewhere, otherwise there will be three meeting points. Therefore only the point of contact is common to the two circles. I claim it is in the line through A, B . This is apparent from §19, if it is shown that it is unique of its relation to A, B . Let there be another one F , if this can happen, and one should have $F.A.B \simeq C.A.B$. Therefore by disjoining, both $F.A \simeq C.A$ and also $F.B \simeq C.B$. Therefore F falls in both circumferences, and thus either they coincide with C , or C is not the only thing in common, which is absurd.

(32) A line and a circle cannot meet each other in more than two points. Let L and M be in line $\overline{X}$ [Figure 12], and also in circumference

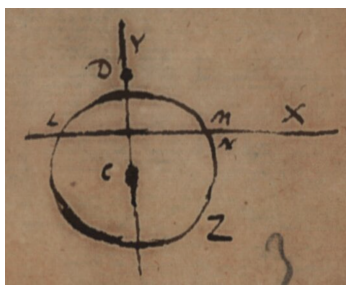


Figure 12

$\overline{Z}$ around C . I say that besides L and M , another point N could not be given. Let D be taken relating in the same way to the line $\overline{X}$ as C

on the other side, by §17. On account of the circle, $L.C \simeq M.C \simeq N.C$ holds. Therefore since the points L, M, N are in a line, they relate in the same way to D as to C , and also $L.D \simeq M.D \simeq N.D$. Therefore by combining, $L.C.D \simeq M.C.D \simeq N.C.D$. Let $\bar{Y}$ be the line through $C.D$; then (by the axiom of §3) $L.\bar{Y} \simeq M.\bar{Y} \simeq N.\bar{Y}$ will hold, i.e. the line $\bar{Y}$ will relate in the same way to the three points L, M, N , which cannot happen by §29.

And so we have revealed the foundations of the straight line and the circle, namely the way these loci can meet each other: a line and a line, a circle and a circle, a line and a circle, by the meetings of which others are determined. From this it follows that the others can also be handled by our calculus.